Unit: 1: System Analysis & Design

Analysis: It emphasizes investigation of the problem and requirements

Design: It emphasizes conceptual solution that fulfills the requirements.

System Analysis vs Design

System Analysis	System Design.
- Investigation of the problems & requirements.	- Emphasizes conceptual solution that fulfills the requirements
- What system must do?	- How system does it?
- Output: requirement documents, analysis model.	- Output: Design documents, database design, architecture UI.
- Nature: Logical & Problem oriented.	- Nature: Technical & solution oriented.
- Activities: Investigation, Interviews	- Activities: Planning system, Designing components.

Information System Analysis & Design

A method used to create and maintain system that performs basic business functions.

→ Main Goal: Improving employee efficiency by applying software solution to key business tasks.

It is structured method and is used to ensure success in business operation.

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Core Concepts of System Analysis & Design

Major Goal: Improving employee efficiency by applying software solution to key business tasks.

System:

A system is an interrelated set of business procedures within one business unit working together for a purpose.

x) Characteristics:

1) Component 2) Environment 3) Boundary
4) Purpose 5) Constraints 6) Interfaces.
7) Input 8) Output.

Software System Development Life Cycle {SDLC}.

SDLC is a standard process followed in an organization to conduct all the steps necessary to build system systemmatically.

It includes of the following phases:

a) Phase 1: System Planning & Selection:

It involves
→ identification of system needs.
→ investigation & determination of scope.

Plan is produced for proposed project and specifying the ~~time~~ required time and resources.

b): Phase 2: System Analysis:

It is the study of current procedures & information systems. Here,
- Determine the requirements.
- Generate alternative designs that meet requirements.
- Recommendation of best alternative.

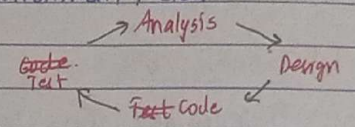
c) Phase 3: System Design:

It is process of converting the recommendation into logical system and then physical system specification.

Logical design focuses on business aspect and how system impacts the existing functional units.

Physical design includes technical specifications like programming languages, database structure, network environment, etc.

The heart of system development is called analysis design implementation.



(d): Phase: 4: System Implementation & Maintainence

System specifications are converted into working system which is then tested and put into use.

The following steps are implemented:

1) Coding 2) Testing
3) Installation 4) User Training
5) Documentation 6) Maintainence.

Roles of the System Analyst:

1) Study problems and needs of organization.
2) Determine the best approach to improve the organization
3) Help system users & managers define requirements for new, improved and enhanced Information system.

Characteristics of Good System Analyst:

a) Analytical skills:

i) Understand the organization and its functions.
ii) Identify problem & opportunities that arise with organization system.
iii) Must be able to analyze & solve problems.
Must implement system ~~skills~~ thinking - seeing organizations and information system as system.

b) Technical Skills

i) Understand the potential and limitation of IT
ii) Must be able to work with programming language, various OS and solve hardware problems.

c) Management skills

i) Managing project, team members, resources, risk, change and time.

D) Interpersonal Skills:

i) Able to work with end users along with other people like analyst & programmers.
ii) ~~They require~~ Able to implement effective oral & written communication skill.

CASE Tools:

Computer Aided Software Engineering are the automated software tools used by system analyst to develop information system.

Eg: 1) UML tools 2) Requirement Management Tools
3) Testing tools 4) Database Design Tools
5) Project Management Tools 6) Code Generation Tools.

Upper case tools are the tools that involve the following tasks:
- Planning - Design - Analysis.

Lower Case Tools are the tools that involve the following tasks:
- Implementation - Testing - Maintainence.

Upper case and lower case tools together form the integrated case tools.

⇒: Merits of CASE tools:

1) Improved productivity: Automates repetitive tasks helping analyst and developers work faster.
Eg: Autogenerating ER-diagram rather than building them.
2) Enhanced Collaboration: Allows multiple team members to work together on same project.
Eg: Use of shared modelling tools to update diagrams & design in real time.
3) Standardization & Consistency: Enforces common formats, rules and documentation standard across different projects.
Eg: UML tools allows us to use same notation and layout for all class diagrams.

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4) Risk Reduction: Helps us to detect errors early reducing project risks.
Eg: Code analysis tools flag potential bugs before the system is tested.

5) Quality Assurance: Ensure higher quality through validation, testing support and documentation checks.
Eg: Testing tools automatically generates test cases to ensure system reliability.

6) Scalability & Reusability: Support large system and allows reusage of components saving time and effort.
Eg: Reusing already designed modules when creating new system.

7) Adaptability & Flexibility: Easy to update designs and requirements as project changes.
Eg: Changes can be made quickly & efficiently using CASE tools.

Demerits of CASE Tools:

1) Initial Cost & Learning Curve: To implement CASE tools, it requires early investment in software licensing, training and infrastructure.

2) Customization complexity: Since CASE tools come with predefined templates, it is difficult the templates ~~as~~ customization required to our needs.

3) Complexity Overhead: CASE tools have many and complex features that may overwhelm users.

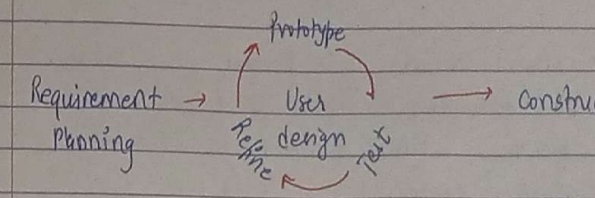
4) Risk of Abandonment: Ineffective integration of CASE tools leads to it being discontinued in use.

Joint Application Design

These are structured meetings where users, managers and analysts work together for several days and system requirements are reviewed.

Through these methods, time and organizational resources are better managed to agree on system requirements and design.

Rapid Application Development:



RAD is an adaptive software development model based on prototyping and quick feedback ~~rather~~ with less emphasis on specific planning.

Developers may require multiple iterations during software development while starting from scratch.

This method ensures that the final outcome is quality-focused and aligns with user interests.

Useful in systems where requirements frequently change throughout development process.

Participatory Design:

Participatory design emphasizes on the role of user community in development process with each user having equal voice in determining system requirements and approving system designs.

In this method, another approach can be to allow elected group of people to control the process.

Emergent Properties:

Emergent properties are consequence of relationships between system components.

It means the properties of whole system when integrated together rather than properties derived from the system components.

Information System Project Management

Definition: IS project management is a process of planning, organizing, leading and controlling resources and activities to achieve specific objectives within context of developing or implementing information system.

Importance: Effective project management ensuring projects are completed on time, within budget, aligning with organizational goals and deliverance of values.

Lifecycle: IS Project Management lifecycle consists the following phases:

1) Initiation:
- Defining project objectives, scope and stake holders.
- Conducting feasibility analysis
- Developing project ~~chato~~ charter.

2) Planning:
- Create project planning, schedule and budget.
- Identifying and allocating resource, roles & responsibilities.

3) Execution:
- Implement project plan, manage tasks and monitor progress
- Addresses issues, communication and ensure quality.

4) Monitoring & Controlling:
- Track performance against the plan.
- Identifying & mitigating risks, managing changes and ensure compliance.

5) Closing:
- Evaluating project outcomes & deliverables.
- Documents lessons, archieve project information.

SMART

SMART refers to criteria for setting goals and objectives.

1) Specific: Should target specific area of improvement or answers specific needs.
2) Measurable: Goal must be quantifiable and measureable progress.
3) Attainable: Goal should be realistic based on resources and constraints.
4) Relevant: Goals should align with other business objectives.
5) Time-bound: Goal must have deadline and defined end.

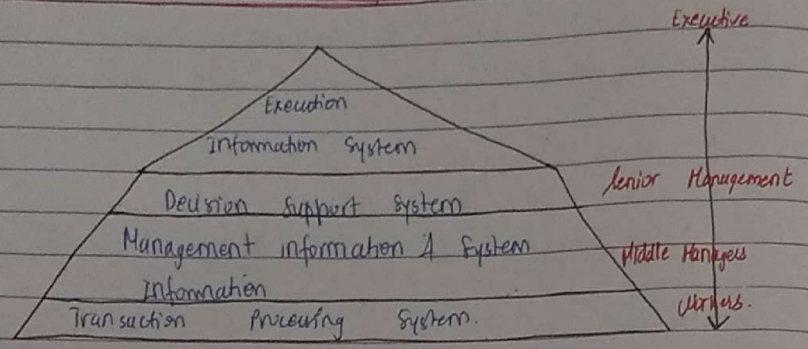
Functional Emergent Properties

These appear when all the parts of system work together to achieve some objective.

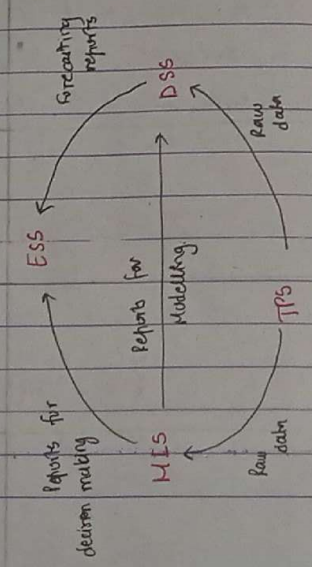
Non-functional Emergent Properties

They relate to the behaviour of system in its operational environment.

Classification of Information System



Systems	Transaction Processing System	Management Information System	Decision Support System	Executive Support System
~~Level~~ Users	~~Bottom~~ Workers.	Middle managers	Upper Analyst ~~Management~~	Executive level
Purpose	Handles daily routine transactions & operations	Provide summarized and structured reports in planning & decision making	Helping non-routine and semi-structured decisions with analytical model.	Provide high level summaries for strategic decision making
Level	Operational level	Middle management	Upper Management	Strategic Management
Example	Payroll system, Order processing	Sales management, inventory control	Financial forecasting & risk analysis tools	Executive dashboards, business intelligence
Function	Records, processes and stores data from day-day activities.	Uses TPS data to generate regular reports.	Combines internal data with external information to support decision making.	Offers trends, forecasts and key performance supporting long term policies



Importance of SAD

1) Requirement Understanding: Ensures developers fully understands what users & businesses absolutely needs.
2) ~~Eg:~~ Better System Quality: Leads to well-structured, efficient and reliable systems.
3) Reduced Development Time & Cost: Detects issues early helping to prevent costly fixes at later time.
4) Improved communication: creates common understanding platform between users, analyst and developers.
5) Enhanced flexibility & adaptability: makes future updates and expansion easier.
6) Better Risk Management: Identifies problems early and provides solutions.
7) Ensures user satisfaction: Produces the system that meet expectations and are easy to use.

& Opposing the importance ie, in negative context gives the channels of not utilizing SDLC ?.

University as a System

System: University.

Purpose: Educating students, Advancing knowledge.

Boundary: Campus, Departments, Faculties, Students, Administration.

Input: Students, Faculties, Administration, Policies, Infrastructure, Technology.

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Purpose: Education, Research, Examination, Evaluation.

Output: Skilled Graduates, Research papers, ~~Trained~~ Trained professionals.

Components: Departments, Administration, Management, IT, Examination office, Transportation.

Interrelationship: All components are interrelated with each other.
Eg: Departments depend on administrative policies, student rely on faculties, etc.

Constraints: Limited budgets & resources, Government regulations, Time constraints.

Interfaces: Communication between departs, faculty and administration

Environment: External factors like educational policies, job markets, technological advancements, etc.

Management Process

Management process includes
- Project Planning
- Resource Allocation
- Process Management.
- Scheduling & Budgeting
- Risk Management

Supporting Process

- Documentation
- Quality Assurance
- Configuration Management
- Testing & training
- Communication

Prototyping

Prototyping involves creating working model of system early in development to show users how the system works.

↔ Advantages

i) Better visualization & interaction models through better understanding.
ii) Reduces misunderstanding & missing requirement.
iii) Speeds up the feedback.

↔ Disadvantages

i) Can give false sense of completeness
ii) Increasing initial development effect.

Guidelines for JAD session

1) Define clear objectives
2) Select the right participants
3) Prepare an agenda
4) Facilitate & Don't dominate.
5) Encourage open Communication
6) Using visual tools & aids.
7) Manage Time Efficiently
8) Document decision & feedback
9) Handles conflicts constructively
10) Summarize & Follow-up

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Agile Methodologies

Agile methodology handles system requirements differently from traditional methodologies.

It focuses on continuous collaboration and iterative development.

The wants and needs of users are collected through uses wants in small, manageable pieces.

A) Iterative & Incremental: Requires evolve through repeated cycles that allows for changes as project programs.

B) Collaboration with Users: Developers, product owners and stakeholders work closely to clarify needs continuously.

c) Priorization: Users stories are prioritized based on business values ensuring most important features are developed first.

D) Flexibility: Agile method accepts that requirements may change and processes are designed to adapt quickly.

E) Documentation: Documentation is light as focus is on ~~heavy~~ working software and communication.

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